

Last Layer KV Cache is All You Need

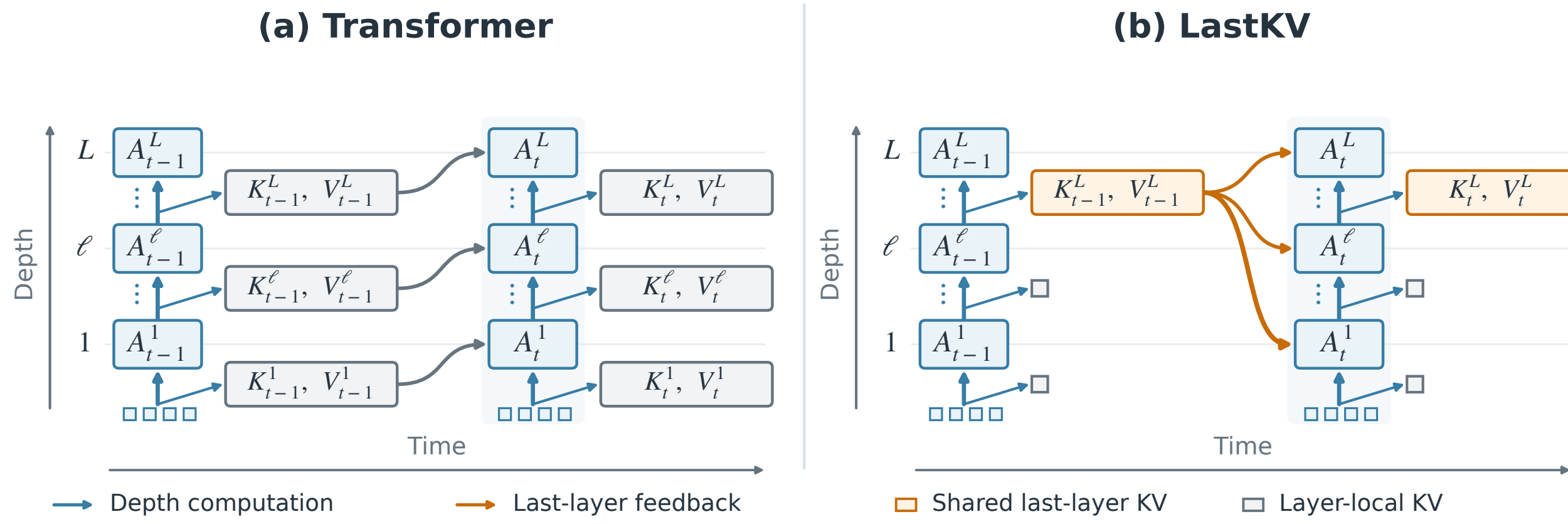
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Recurrence across time and depth

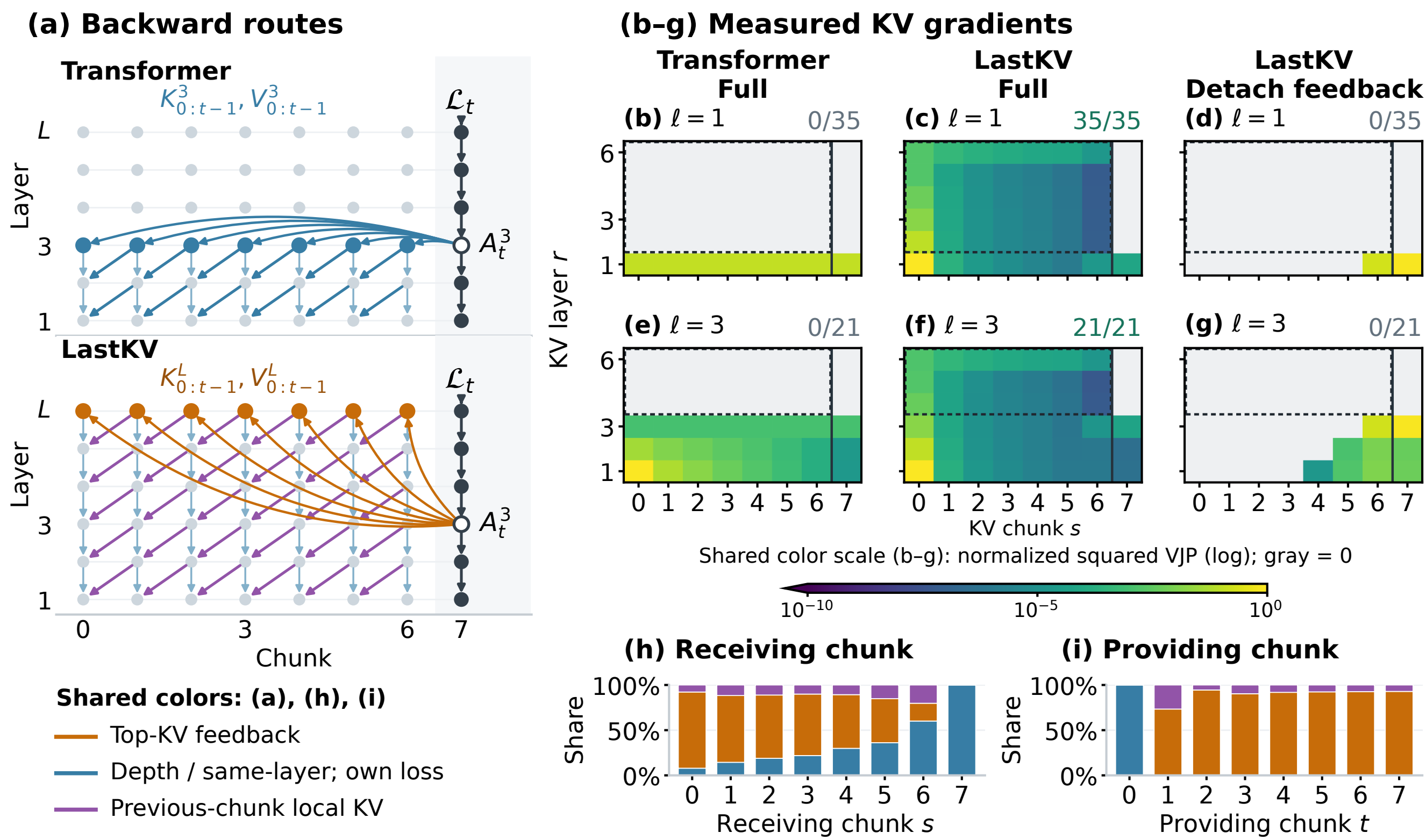
1 Deep past states feed every future layer



Training: parallel within chunks, recurrent across chunks.

Older history uses shared last-layer KV. The previous chunk retains blended local KV.

3 Differentiable recurrence across time and depth

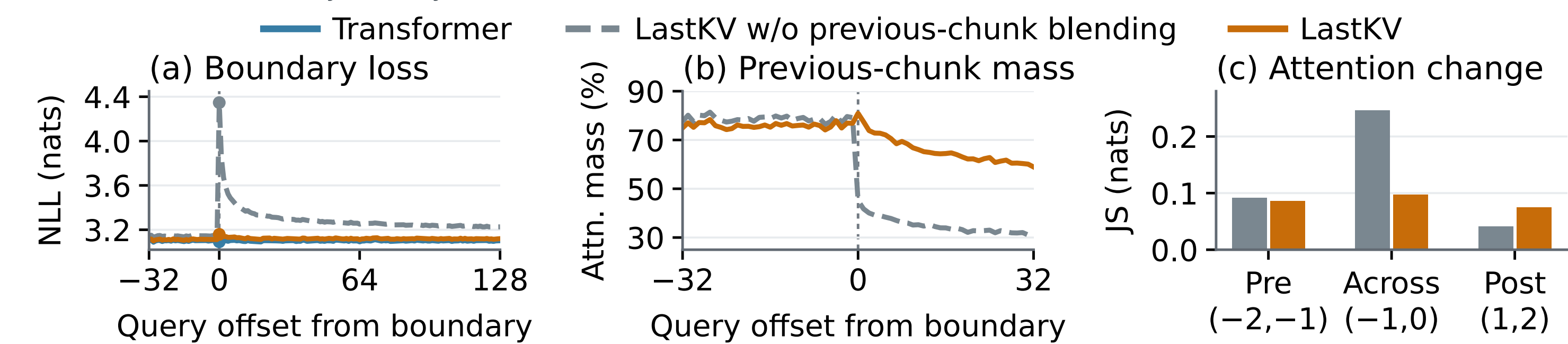


Deep historical KV receive recurrent gradients.

Random initialization: 6 layers, T4K, C512, 3 seeds.
Detached feedback keeps forward KV, but removes higher-layer gradient paths.

5 Chunk recurrence and boundary blending

The previous chunk blends local and last-layer KV.
Older chunks retain only last-layer KV.

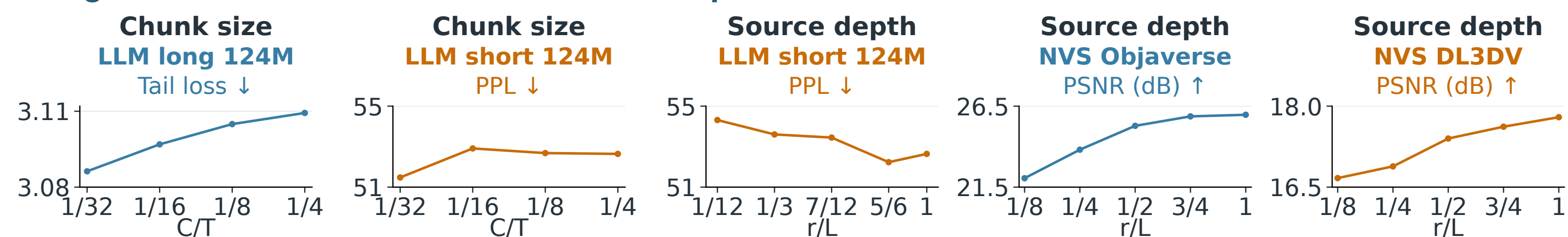


Task	Metric	Without blending	LastKV
LLM long (124M)	Tail loss ↓	3.141	3.102
LLM long (124M)	RULER avg. ↑	12.17	17.52
LLM short (124M)	WikiText PPL ↓	52.838	52.642
LLM short (124M)	Ten-task avg. ↑	42.33	42.54

Blending raises RULER by 5.35 points.

Both variants retain recurrence. 124M with direct last-layer KV reuse.
Boundary diagnostics compare separate checkpoints. Attention uses four matched windows.

Design ablations: chunk size and source depth



C/T: chunk ratio. Blending window changes with C. Source depth: r/L.
LLM PPL is non-monotonic near the last layer. Short runs use H100/H200.

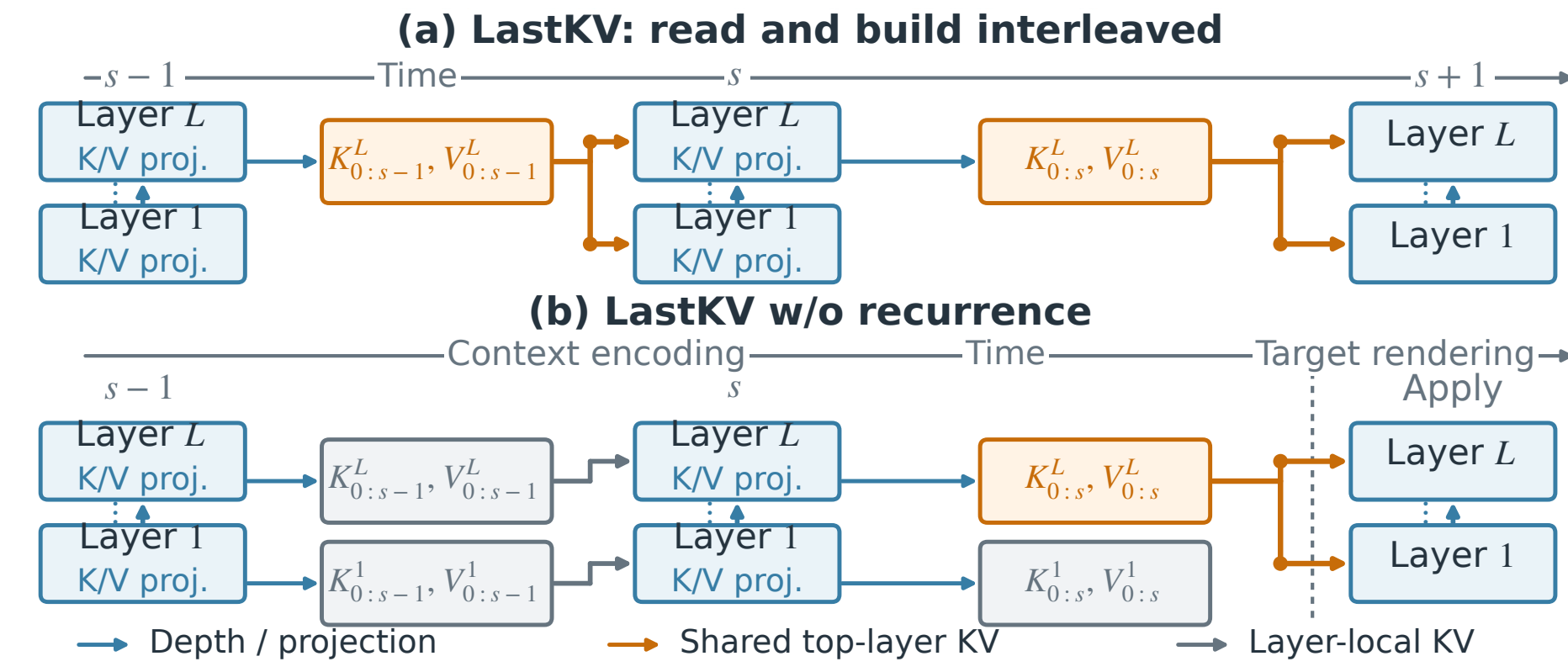
2 State tracking and memory retention

Method	Fixed FSA	Mode switch	State tracking		Perm. comp.	Memory retention	
			Gen. mod-sum	Group word		MQAR	Selective copy
RNN	33.0	50.6	2.0	19.3	12.7	0.2	0.2
Transformer	40.7	78.8	11.4	25.3	57.7	68.4	99.9
SWA	40.3	65.8	12.9	38.8	46.2	37.2	71.2
Mamba-2	40.3	73.0	5.9	80.0	43.8	100.0	0.5
DeltaNet	33.3	72.2	5.1	48.9	55.6	100.0	84.8
Gated DeltaNet	38.8	73.8	4.2	59.0	56.6	100.0	80.7
Kimi Delta Attention	42.2	75.7	3.8	49.7	57.9	100.0	92.8
LaCT	42.1	88.9	25.6	27.6	59.5	100.0	99.8
LastKV	43.2	92.6	21.3	27.6	59.8	100.0	100.0

5/7 best or tied-best; all 7 above Transformer.

Token accuracy (%), †. 2K context, C512. Means over task settings.

4 Recurrence ablation



Both targets read last-layer KV.
The control builds context with layer-local attention.

Recurrence ablation results

Task	Metric	Without recurrence	LastKV
Video (1.3B)	VBench total ↑	81.36	81.90
NVS: Objaverse	PSNR (dB) ↑	25.31	25.97
NVS: DL3DV	PSNR (dB) ↑	17.32	17.80
LLM short (124M)	WikiText PPL ↓	53.22	52.64
LLM short (124M)	Ten-task avg. ↑	42.47	42.54
LLM long (124M)	Tail loss ↓	3.10	3.10
LLM long (124M)	RULER avg. ↑	9.43	17.52

124M RULER +8.09 points. NVS +0.66 / +0.48 dB.

Ablation settings differ from the main evaluation below.

Main evaluation: language, 3D and video

8B CPT: HumanEval +13.84, 17-benchmark avg. +2.41 points.

8B model	MBPP	MBPP+*	HumanEval	HumanEval+	GPQA-D	GSM8K	MATH-500	Avg.†
Start	15.10	3.35	10.87	7.77	22.64	20.78	7.20	26.60
Transformer	49.70	75.32	36.46	30.39	27.63	71.24	56.70	46.21
LastKV	50.60	74.27	50.30	44.63	32.45	71.12	59.70	48.62

Scores (%), †. * Restricted-test MBPP+. GPQA-D uses CoT. † Equal-weight mean of 17 benchmarks.

Main evaluation: recurrent language models

Method	124M				760M			
	Short PPL ↓	Short avg. †	Long PTL ↓	Long RULER †	Short PPL ↓	Short avg. †	Long PTL ↓	Long RULER †
Transformer	54.725	41.5	3.0869	16.52	26.565	48.5	2.598	34.2
T ² MLR-LastKV	57.841	40.6	—	—	28.071	48.3	—	—
FBT-LastKV	57.205	40.9	—	—	29.741	46.9	—	—
LastKV	52.642	42.5	3.1067	20.05	25.912	49.9	2.556	34.3

Short: WikiText PPL / ten-task avg. Long: 30K+ tail loss / mean RULER. Selected methods.
T²MLR/FBT: Jacobi recurrence adapted to last-layer KV sharing. —: not reported.

Novel view synthesis: PSNR (dB), †

Dataset	Transformer	LastKV
Objaverse (GSO)	25.36	25.94†
DL3DV	17.41	17.80

† Objaverse: LastKV uses in-training random views, others fixed views.
DL3DV uses matched fixed views. Curves also compare LaCT.

Video: VBench (%) after teacher-forced fine-tuning

Size / model	Quality †	Semantic †	Total †	Votes
1.3B Transformer	83.1	76.5	81.8	85
1.3B LastKV	83.5	78.0	82.4	355
14B Transformer	82.2	72.7	80.3	114
14B LastKV	83.0	76.4	81.6	326

